

Labsheet_2

1. **Student Registration Form: Create a Student Registration Form using HTML. Include text fields, radio buttons, checkboxes, dropdown list, textarea, submit, and reset buttons. Apply CSS to style the form.**

Student Registration Form

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
    }

    .form-container {
      width: 450px;
      margin: 40px auto;
      padding: 25px;
      background-color: white;
      border-radius: 10px;
      box-shadow: 0 0 10px gray;
    }

    h2 {
      text-align: center;
      color: #333;
    }

    label {
      font-weight: bold;
    }

    input, select, textarea {
      width: 100%;
      padding: 8px;
```

```
        margin: 8px 0 15px;
        box-sizing: border-box;
    }

    input[type="radio"],
    input[type="checkbox"] {
        width: auto;
        margin: 5px;
    }

    textarea {
        height: 80px;
        resize: none;
    }

    .buttons {
        text-align: center;
    }

    input[type="submit"],
    input[type="reset"] {
        width: 100px;
        padding: 10px;
        margin: 5px;
        border: none;
        border-radius: 5px;
        cursor: pointer;
    }

    input[type="submit"] {
        background-color: green;
        color: white;
    }

    input[type="reset"] {
        background-color: red;
        color: white;
    }
</style>
</head>
```

```
<body>
```

```
<div class="form-container">
```

```
<h2>Student Registration Form</h2>
```

```
<form>
```

```
<label>Full Name:</label>
```

```
<input type="text" name="name" placeholder="Enter your name">
```

```
<label>Email:</label>
```

```
<input type="email" name="email" placeholder="Enter your email">
```

```
<label>Password:</label>
```

```
<input type="password" name="password" placeholder="Enter password">
```

```
<label>Gender:</label><br>
```

```
<input type="radio" name="gender" value="male"> Male
```

```
<input type="radio" name="gender" value="female"> Female
```

```
<input type="radio" name="gender" value="other"> Other
```

```
<br><br>
```

```
<label>Course:</label>
```

```
<select name="course">
```

```
<option value="">Select Course</option>
```

```
<option value="bca">BCA</option>
```

```
<option value="btech">B.Tech</option>
```

```
<option value="bba">BBA</option>
```

```
<option value="mca">MCA</option>
```

```
</select>
```

```
<label>Hobbies:</label><br>
```

```
<input type="checkbox" name="hobby" value="reading"> Reading
```

```
<input type="checkbox" name="hobby" value="sports"> Sports
```

```
<input type="checkbox" name="hobby" value="music"> Music
```

```
<input type="checkbox" name="hobby" value="coding"> Coding
```

```
<br><br>
```

```

        <label>Address:</label>
        <textarea name="address" placeholder="Enter your address"></textarea>

        <div class="buttons">
            <input type="submit" value="Submit">
            <input type="reset" value="Reset">
        </div>

    </form>

</div>

</body>
</html>

```

Output

The webpage displays a styled Student Registration Form containing:

- Full Name
- Email
- Password
- Gender — Radio buttons
- Course — Dropdown list
- Hobbies — Checkboxes
- Address — Textarea
- Submit button
- Reset button

2.College Website using Frames: Create a college website using HTML Frames with Header, Navigation Menu, and Main Content sections. Add navigation links that display different pages in the main frame.

Ko</html>

2. "header.html"

```

<!DOCTYPE html>
<html>
<head>

```

```
<title>Header</title>
</head>

<body style="background-color: lightblue; text-align:center;">
  <h1>COER University</h1>
  <h3>College of Smart Computing</h3>
</body>
</html>
```

3. "menu.html"

```
<!DOCTYPE html>
<html>
<head>
  <title>Navigation Menu</title>
</head>

<body style="background-color: lightgray;">

  <h3>Navigation Menu</h3>

  <a href="home.html" target="main">Home</a><br><br>

  <a href="about.html" target="main">About Us</a><br><br>

  <a href="courses.html" target="main">Courses</a><br><br>

  <a href="contact.html" target="main">Contact Us</a><br><br>

</body>
</html>
```

4. "home.html"

```
<!DOCTYPE html>
<html>
<body>

  <h2>Welcome to COER University</h2>
```

```
<p>
  Welcome to our college website.
  We provide quality education and practical
  learning opportunities to students.
</p>
```

```
</body>
</html>
```

5. "about.html"

```
<!DOCTYPE html>
<html>
<body>
```

```
  <h2>About Us</h2>
```

```
  <p>
    COER University is committed to providing
    quality education, innovation and
    skill development.
  </p>
```

```
</body>
</html>
```

6. "courses.html"

```
<!DOCTYPE html>
<html>
<body>
```

```
  <h2>Courses Offered</h2>
```

```
  <ul>
    <li>BCA</li>
    <li>B.Tech</li>
    <li>BBA</li>
    <li>MCA</li>
  </ul>
```

```
</body>
</html>
```

7. "contact.html"

```
<!DOCTYPE html>
<html>
<body>

    <h2>Contact Us</h2>

    <p>Email: info@coeruniversity.ac.in</p>
    <p>Phone: +91-XXXXXXXXXX</p>
    <p>Address: Roorkee, Uttarakhand</p>

</body>
</html>
```

Output

The website is divided into three sections:

Header → Displays college name and title.

Navigation Menu → Contains links:

- Home
- About Us
- Courses
- Contact Us

Main Content → Displays the selected page when a navigation link is clicked.

3.Login and Admission Form: Design a Login Form and College Admission Form using HTML. Use appropriate form elements and apply CSS for formatting, borders, spacing, fonts, and buttons.

Login and College Admission Form

1. Login Form

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
    }

    .login-box {
      width: 350px;
      margin: 60px auto;
      padding: 25px;
      background-color: white;
      border: 1px solid #ccc;
      border-radius: 10px;
      box-shadow: 0 0 10px gray;
    }

    h2 {
      text-align: center;
    }

    label {
      font-weight: bold;
    }

    input {
      width: 100%;
      padding: 10px;
      margin: 8px 0 15px;
      box-sizing: border-box;
    }

    button {
      width: 100%;
      padding: 10px;
```

```
        background-color: green;
        color: white;
        border: none;
        border-radius: 5px;
        font-size: 16px;
    }

    button:hover {
        background-color: darkgreen;
    }
</style>
</head>

<body>

<div class="login-box">

    <h2>Login Form</h2>

    <form>
        <label>Username:</label>
        <input type="text" placeholder="Enter Username">

        <label>Password:</label>
        <input type="password" placeholder="Enter Password">

        <button type="submit">Login</button>
    </form>

</div>

</body>
</html>
```

2. College Admission Form

```
<!DOCTYPE html>
<html>
<head>
    <title>College Admission Form</title>
```

```
<style>
  body {
    font-family: Arial, sans-serif;
    background-color: #eef2f3;
  }

  .admission-form {
    width: 600px;
    margin: 30px auto;
    padding: 25px;
    background-color: white;
    border: 2px solid #555;
    border-radius: 10px;
  }

  h1 {
    text-align: center;
    font-size: 28px;
  }

  label {
    font-weight: bold;
  }

  input, select, textarea {
    width: 100%;
    padding: 9px;
    margin: 7px 0 15px;
    border: 1px solid #aaa;
    border-radius: 5px;
    box-sizing: border-box;
  }

  input[type="radio"],
  input[type="checkbox"] {
    width: auto;
    margin: 5px;
  }
```

```
    textarea {
      height: 80px;
    }

    .buttons {
      text-align: center;
    }

    input[type="submit"],
    input[type="reset"] {
      width: 120px;
      color: white;
      border: none;
      cursor: pointer;
      margin: 5px;
    }

    input[type="submit"] {
      background-color: green;
    }

    input[type="reset"] {
      background-color: red;
    }
  </style>
</head>

<body>

<div class="admission-form">

  <h1>College Admission Form</h1>

  <form>

    <label>Student Name:</label>
    <input type="text" placeholder="Enter your name">

    <label>Email:</label>
    <input type="email" placeholder="Enter your email">
```

```
<label>Mobile Number:</label>
<input type="tel" placeholder="Enter mobile number">
```

```
<label>Date of Birth:</label>
<input type="date">
```

```
<label>Gender:</label><br>
<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
<input type="radio" name="gender"> Other
<br><br>
```

```
<label>Select Course:</label>
<select>
  <option>Select Course</option>
  <option>BCA</option>
  <option>B.Tech</option>
  <option>BBA</option>
  <option>MCA</option>
</select>
```

```
<label>Hobbies:</label><br>
<input type="checkbox"> Reading
<input type="checkbox"> Sports
<input type="checkbox"> Music
<input type="checkbox"> Coding
<br><br>
```

```
<label>Address:</label>
<textarea placeholder="Enter your address"></textarea>
```

```
<label>Qualification:</label>
<input type="text" placeholder="Enter your qualification">
```

```
<div class="buttons">
  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
</div>
```

```
</form>
```

```
</div>
```

```
</body>
```

```
</html>
```

Output

Login Form:

- Username
- Password
- Login button

College Admission Form:

- Student Name
- Email
- Mobile Number
- Date of Birth
- Gender
- Course
- Hobbies
- Address
- Qualification
- Submit and Reset buttons

CSS is used for borders, spacing, fonts, background, form size, buttons, and alignment.

4. JavaScript Calculator: Create a simple calculator using HTML and JavaScript to perform addition, subtraction, multiplication, and division of two numbers.

JavaScript Calculator

Program

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Simple Calculator</title>

<style>
  body {
    font-family: Arial, sans-serif;
    text-align: center;
    background-color: #f2f2f2;
  }

  .calculator {
    width: 350px;
    margin: 50px auto;
    padding: 25px;
    background-color: white;
    border: 2px solid #333;
    border-radius: 10px;
  }

  input {
    width: 90%;
    padding: 10px;
    margin: 10px;
    font-size: 16px;
  }

  button {
    padding: 10px 15px;
    margin: 5px;
    font-size: 16px;
    cursor: pointer;
  }

  #result {
    font-weight: bold;
    font-size: 20px;
    margin-top: 15px;
  }
</style>
</head>
```

```
<body>

<div class="calculator">

  <h2>Simple Calculator</h2>

  <input type="number" id="num1" placeholder="Enter first number">

  <input type="number" id="num2" placeholder="Enter second number">

  <br>

  <button onclick="calculate('+')">+</button>
  <button onclick="calculate('-')">-</button>
  <button onclick="calculate('*')">×</button>
  <button onclick="calculate('/')">÷</button>

  <p id="result">Result: </p>

</div>

<script>
  function calculate(operator) {

    let num1 = parseFloat(document.getElementById("num1").value);
    let num2 = parseFloat(document.getElementById("num2").value);
    let result;

    if (isNaN(num1) || isNaN(num2)) {
      document.getElementById("result").innerHTML =
        "Please enter both numbers.";
      return;
    }

    if (operator == "+") {
      result = num1 + num2;
    }
    else if (operator == "-") {
      result = num1 - num2;
    }
  }
}
```

```
        else if (operator == "*") {
            result = num1 * num2;
        }
        else if (operator == "/") {
            if (num2 == 0) {
                result = "Cannot divide by zero";
            } else {
                result = num1 / num2;
            }
        }
    }

    document.getElementById("result").innerHTML =
        "Result: " + result;
}
</script>

</body>
</html>
```

Example Output

If:

First Number: "20"

Second Number: "5"

- Addition → "25"
- Subtraction → "15"
- Multiplication → "100"
- Division → "4"

Result

The calculator successfully performs addition, subtraction, multiplication, and division of two numbers using JavaScript.

5. JavaScript Validation: Create a Student Registration Form using HTML and JavaScript. Validate fields such as Name, Email, Mobile Number, Password, and Age before submitting the form. Display appropriate validation messages.

JavaScript Validation – Student Registration Form

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
    }

    .container {
      width: 450px;
      margin: 40px auto;
      padding: 25px;
      background-color: white;
      border: 2px solid #555;
      border-radius: 10px;
    }

    h2 {
      text-align: center;
    }

    label {
      font-weight: bold;
    }

    input {
      width: 100%;
      padding: 9px;
      margin: 7px 0;
      box-sizing: border-box;
      border: 1px solid #aaa;
      border-radius: 5px;
    }

    .error {
```

```
        color: red;
        font-size: 14px;
    }

    button {
        padding: 10px 20px;
        margin-top: 15px;
        background-color: green;
        color: white;
        border: none;
        border-radius: 5px;
        cursor: pointer;
    }
</style>
</head>

<body>

<div class="container">

    <h2>Student Registration Form</h2>

    <form onsubmit="return validateForm()">

        <label>Name:</label>
        <input type="text" id="name">
        <div id="nameError" class="error"></div>

        <label>Email:</label>
        <input type="text" id="email">
        <div id="emailError" class="error"></div>

        <label>Mobile Number:</label>
        <input type="text" id="mobile">
        <div id="mobileError" class="error"></div>

        <label>Password:</label>
        <input type="password" id="password">
        <div id="passwordError" class="error"></div>
```

```
<label>Age:</label>
<input type="number" id="age">
<div id="ageError" class="error"></div>

<button type="submit">Register</button>
```

```
</form>
```

```
</div>
```

```
<script>
```

```
function validateForm() {
```

```
    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
    let mobile = document.getElementById("mobile").value.trim();
    let password = document.getElementById("password").value;
    let age = document.getElementById("age").value;
```

```
    let valid = true;
```

```
    // Clear previous messages
```

```
    document.getElementById("nameError").innerHTML = "";
    document.getElementById("emailError").innerHTML = "";
    document.getElementById("mobileError").innerHTML = "";
    document.getElementById("passwordError").innerHTML = "";
    document.getElementById("ageError").innerHTML = "";
```

```
    // Name validation
```

```
    if (name == "") {
        document.getElementById("nameError").innerHTML =
            "Name is required.";
        valid = false;
    }
```

```
    // Email validation
```

```
    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
    if (email == "") {
```

```

    document.getElementById("emailError").innerHTML =
        "Email is required.";
    valid = false;
}
else if (!emailPattern.test(email)) {
    document.getElementById("emailError").innerHTML =
        "Enter a valid email address.";
    valid = false;
}

// Mobile validation
let mobilePattern = /^[0-9]{10}$/;

if (mobile == "") {
    document.getElementById("mobileError").innerHTML =
        "Mobile number is required.";
    valid = false;
}
else if (!mobilePattern.test(mobile)) {
    document.getElementById("mobileError").innerHTML =
        "Mobile number must contain 10 digits.";
    valid = false;
}

// Password validation
if (password == "") {
    document.getElementById("passwordError").innerHTML =
        "Password is required.";
    valid = false;
}
else if (password.length < 6) {
    document.getElementById("passwordError").innerHTML =
        "Password must be at least 6 characters.";
    valid = false;
}

// Age validation
if (age == "") {
    document.getElementById("ageError").innerHTML =
        "Age is required.";

```

```
        valid = false;
    }
    else if (age < 18 || age > 60) {
        document.getElementById("ageError").innerHTML =
            "Age must be between 18 and 60.";
        valid = false;
    }

    if (valid) {
        alert("Registration successful!");
    }

    return valid;
}

</script>

</body>
</html>
```

Validation Rules

Field| Validation

Name| Cannot be empty

Email| Must be in valid email format

Mobile Number| Must contain exactly 10 digits

Password| Minimum 6 characters

Age| Must be between 18 and 60

Output

If any field is invalid, an appropriate error message in red is displayed below that field. If all fields are valid, the message "Registration successful!" is displayed.

